

Nishchay Agarwal

Software Development Engineer | Backend | GenAI

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PROFILE

Software Development Engineer with 4+ years of experience in designing, developing, and maintaining scalable backend systems, RESTful APIs, and cloud-native applications. Strong expertise in Python, FastAPI, Flask, and Django, with hands-on experience across AWS and Azure cloud platforms. Proven ability to design and deploy RAG-based GenAI solutions, integrate LLMs with enterprise data, and deliver secure, production-ready software. Adept at writing clean, testable code, optimizing performance, and collaborating across cross-functional teams in agile environments.

EXPERIENCE

- Rockwell Automation** Bengaluru, India
Software Development Engineer II Feb 2023 – Present
 - GenAI & RAG Systems:** Designed and developed end-to-end RAG-based AI agents using structured and unstructured enterprise data, enabling accurate, context-aware responses through LLM integration.
 - Knowledge Ingestion:** Built scalable document ingestion and vector search pipelines (semantic chunking, embeddings, pgvector/FAISS) and optimized retrieval using hybrid search and prompt engineering.
 - LLM Integration:** Developed the FactoryTalk Energy Manager – Private Cloud, building high-throughput data pipelines with Python, Redis, MongoDB, and KairosDB for real-time and historical energy analytics.
 - Backend APIs:** Ensured performance, security, and high availability of GenAI services and cloud platforms while collaborating closely with product and data teams.
- ZS Consultancy (Ocean CDR – Contract)** Remote
Software Development Engineer May 2022 – Feb 2023
 - Backend Services:** Built Python-based backend services and REST APIs for large-scale data ingestion systems.
 - ETL Automation:** Automated ETL workflows, reducing manual data processing effort by ~30%
 - Cloud Monitoring:** Integrated AWS CloudWatch and FSx for system monitoring and observability.
- Knowledge Lens** Bengaluru, India
Software Development Engineer Dec 2021 – May 2022
 - API Development:** Developed RESTful APIs using FastAPI and Django to automate recruitment workflows and backend operations.
 - Object Storage:** Integrated MinIO for secure document and image storage and retrieval.
 - Testing:** Implemented unit tests using pytest, improving code reliability and reducing post-deployment issues by 20%.

EDUCATION

- PGDM in Software Engineering for Data Science** Hyderabad, India
IIIT Hyderabad (in collaboration with Great Learning) Mar 2021 – Apr 2022
 - Coursework:** Software Development, Data Management, and Algorithm Design.
 - Automation Project:** Automated Active Directory (AD) group access requests to Databricks using internal and third-party tools.
- Bachelor of Technology (B.Tech) – Mechanical Engineering** Roorkee, India
College of Engineering Roorkee Jun 2016 – Sep 2020
 - Tools:** SolidWorks, CATIA, and Finite Element Analysis (FEA) using ANSYS.

TECHNICAL SKILLS

- Backend & Software Engineering:** Software Development & Backend Engineering, Microservices, Distributed Systems
- Programming Languages:** Python, Databases (SQL, PostgreSQL, MongoDB, RedisDB)
- API Development:** RESTful APIs (FastAPI, Flask, Django)
- Generative AI:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering
- Databases & Search:** Vector Databases, Semantic Search
- Cloud Platforms:** AWS, Azure
- Security:** API Security, OAuth2, JWT
- Testing & CI/CD:** pytest, Continuous Integration/Deployment.
- DevOps & Tools:** Docker, Linux, Git